

ActInf MorphStream 003.1 ~ Anand Subramoney

"Principles of scalability and biological inspirations"

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[Laughter] so hello everybody oh can you just go on mute if you're not speaking sorry I'm just going to do a bit of an introduction um so welcome to the third um morph stream series the others are on YouTube if you've not had a chance to catch those so just a reminder of the purpose of these meetings are to educate and Inspire at the intersection of active inference and neuromorphic Computing um if you've not met me before I'm Sarah Hamburg so I'm predominantly a neuroscientist I did my PhD neuroscience at UCR which is where I first came across active inference um and I've just finished a postto in neuromor it Computing and I'm part of the Sab for the active inference Institute and I've think I've been involved with them since about 2020 when I was trying to work out earlier but I'm really pleased today that we're joined by Professor Annan sub money so he's an assistant professor in the Department of computer science at Royal Holloway London he's broadly interested in learning and intelligence so that's both algorithmic and biological his current research is focusing on understanding the principles of scalability in deep learning which is what he'll be talking about today sorry I'm just admitting some people um so he aims to use principles of scalability to build models that are more efficient and can scale up seamlessly his research draws inspiration from neuroscience and biology in his Quest To Build a Better and more General artificial intelligence and you can visit his website onand subber money for more information I'll pop that in the chat as well so his talk today is on principles of scalability and biological Inspirations he's going to talk about how current models scale and what we can learn from the efficiency of biological brains one of the central themes will be sparsity its significant role in scalable systems and its synergies with neuromorphic Hardware he'll present existing ideas based on spiking networks and recent work from his group which is focused on using various forms of sparity and distributed learning to improve the scalability and efficiency of our learning models so the format will be that the first part um arnand will speak and then we'll sort of have a more General open conversation between all of us um arand I don't know if you

want people to interrupt you with questions or not if you just want to take them at the end that's entirely up to you yeah I'm actually completely fine with taking questions during the talk cool okay right well without further ado uh yeah we'd love to hear your presentation yeah thank you Sarah and thank you very much for the invitation uh so yeah like Sarah mentioned I'm going to be talking about principles of scalability and biological Inspirations uh so I've been working with computation neuroscience and machine learning for quite a long time and have been thinking about uh well I mean how intelligence Works to some extent but also really uh a big part of it is about understanding how various how biological models scale and by extension how artificial intelligence model scale and that's what I'm going to be talking about today so just to start off with uh things that we already know I mean so AI is really starting to become better than humans at lots of things right so this is kind of like a plot of performance of various AI models in relation to humans and you see that handwriting recognition and speech recognition image recognition so this all of these have been like improving very very fast uh but recently things like language understanding and reading comparison have also in imp improved and become better than humans so there is really some kind of like a major movement going on in AI where things are really becoming uh better very fast and so the question is what is happening or why does it so work so well so there have been a lot of uh advances in the past do decades uh because we've started using deep neural networks with lots of layers and lots of parameters and we have uh as as a the machine Learning Community has figured out how to train these models well and get them to work well so there has been algorithmic advances in stochastic gradient descent using ReLU layer skip connections normalization layers and all of these have been like very important in giving us the ability to train these very deep networks the other aspect uh that's things the other thing that has changed in the past two decades is that we have lots more data right so we have access to a lot more data so uh which helps a lot with supervised learning but then we've also uh as a community figured out how self-supervised learning can be used to train these models and get them to do interesting things and the third thing is we have a lot more compute and I will talk about this so this is uh actually going to be one of the focus of my talk uh and we also have scalable architectures right so even if you have lots of compute it doesn't really help if uh the machine learning architectures that we have don't scale and make use of the compute that we have so that's obviously like a very closely related and important part so just to look at compute itself so this is kind of like a plot of various models all the way from 1985 to modern day and so what's interesting here is I mean first of all uh before GPU Computing which is this

left side of the plot over here uh what you see is that um the computer has been doubling every 24 months so this has been just standard M law we' have been getting faster and faster processes but there is this sort of discontinuity where people figured out that they could use gpus that were originally designed for for graphics uh they could use that to do machine learning and so what you see here is that the slope of the uh increase in per in amount of compute available changes quite significantly it becomes a lot more Steep and so here the compute starts doubling every 3.4 months starting about 2012 or so and then the third major aspect is these set of models uh that we're familiar with now the Transformer models and you see that uh the comput has been doubling every two months uh with with these models so uh so it looks like compute obviously plays a very very major role in the AI capabilities that we have but also the models that are able to take advantage of this computer so what happened since 2018 which is what uh uh which is when uh this comput started doubling every two months so one of the major advances in 2018 was uh a new scalable deep learning architectures that were published uh that actually was able to take use of this or make use of the compute very well and then a lot of engineering effort and um money spent uh actually getting these models uh scaling up these models built on these architectures right actually making use of the computer and this is kind of uh always a separate concern because even if you have a great model that scales really well there needs to be like a lot of engineering done to actually scale it up and the scalable model that was published in 2018 was the Transformers so most of you might have seen this already so this is the paper attention is all you need which was published in two first published in 2017 and has already gotten tens of thousands of citations if not more close to 100,000 cations and uh this is an architecture that replaced lstms which were the dominant recurrent Network architecture for sequence processing at the time so um so there are various reasons why Transformers are scalable uh and what was kind of interesting is that so they're scalable not just in the ability to be able to run them across a lot of gpus or to use up a lot of compute but uh the important aspect of scaling up over here is that as you spent more compute uh the model performance improved right and that's that's a really important aspect and this is kind of the scaling plot from the now famous uh gpt3 paper and what you see is uh on the x-axis is the amount of compute that was used to train the model so this kind of includes a combination of the model size and the uh length of the training which is also like very closely related to the amount of data that was available and uh the y axis here is the validation error and and what you see here is that as you spend more and more compute as you have larger models and as you train them for longer uh with with certain amounts of data

the validation error goes down and uh so uh you can you can really like invest comp and get returns out of using using a poet comput and here there's like a power law relationship between these two and depending upon the model more recently people have observed there's like a much more complicated relationship but these models do tend to improve in performance as you invest more compute in them uh up till up to a certain limit obviously okay so why do these uh architecture scale well is I mean obviously one of the main questions and then by extension can we scale these models even better and how do we do this so to answer the question of why do Transformers scale so well I mean there are lots of reasons I'm just like focusing on a few of them but uh like I said Transformers were uh proposed as a replacement to recurrent networks in the form of lstms and they use something called self attention to process a quences so U what they did was they replaced recurrent networks which are known to be hard to train with feed forward networks that are easier to train and uh the other kind of very important U advantage of Transformers was that they allow parallelization across the length of the sequence right and so when you have like really long sequences or I mean any number of sequences in any length of sequence the recur Network you always have to uh process it one SE sequence element at a time and with Transformers you could process the entire length of a sequence in one go because of the particular way the architecture is designed and that's a pretty huge Advantage because now you have a whole new dimension of parallelization and um so the second reason why Transformers scale so well is because uh we already figured out that gpus work well for machine learning and Transformers were really really uh great fit for gpus right so you could a Transformer and a GPU and really make full use of the compute provided by the GPU I mean we don't know if uh Transformers are intrinsically better as a model or if it is just because of the good fit between Transformers and the available compute U the available types of compute that we have today so gpus primarily so uh which is which is kind of this uh question of was it a hardware Lottery right um so a hardware Lottery is used to describe when a research idea wins because it is suited to the available software and Hardware better and not because the idea is superior to the alternative research directions and so for Transformers it's not fully clear if it was just a hardware lottery or if there's like something uh very intrinsically advantageous about it okay so so so the scaling of Transformers I mean I'm not going to go into any further Det because there's a lot of uh literature published on this but now the question is can we scale even better than Transformers and this is kind of a plot U uh taken from this paper by uh boan who's a professor in neuromorphic professor in trans Stanford and uh so this is a plot of the uh

on the x-axis you see the number of neurons and y axis the energy consumption right and uh so this Gray Line over here is where we are at currently so that's kind of the standard silicon chips uh in 2D and U the biological uh systems that we know of so biological brains is this darker line over here which kind of uh you see has a much steeper slope of scaling compared to the Silicon chips that we have so I'll talk about what this line in between is in a moment but uh because of just like very practical reasons just physical constraints and the sizes of things and the amount of energy that it can consume and just the proportion of things like proportion of wiring uh versus the amount of compute that you have available so for various different reasons you really want the steeper slope uh in both the algorithms and the hardware that you have because otherwise it makes it like close to impossible to achieve aieve the same level of scale so over here biological brains have reached like 86 billion neurons and uh well I mean this is a little bit uh older but this because of the steep slope if you wanted to reach the same number of units you're going to consume quite a bit more energy right so Transformers are probably like somewhere up there and biological brains are over here and at some point this becomes a deal breaker so you can this this starts like uh providing like an uh upper limit on how big you can scale the models so Energy Efficiency in that sense is like uh like a very fundamentally important uh aspect u i mean assuming that scaling models always gives you this U dividend of better performance then uh it it at some point without efficient models it becomes impossible to reach sizes that can do interesting things okay so uh what does biology do how does biology scale because we see that it has this steep slope of scaling and one of the things that it does is um it uses a lot of sparsity I mean a lot of it is also again like just because of physical limits and the amount of uh cells the density of the cells that you can have the amount of energy available to the brain and so on but mean since this has evolved under these constraints what we have ended up with are uh is a substrate for computation that is extremely sparse both in time and space so uh this is kind of like a plot of What U the neuron activity looks like uh for for like one of these typical neurons and what you usually see is that there are these kind of spikes so there are times when the neuron decides to communicate with the rest of the neurons and then most of the rest of the time it's really not doing much and then so you have like these spikes of outputs and this is sparsity in time so um in terms of activity and then you also have sparsity in terms of space which is just the connectivity of the neuron is very very sparse so not all neurons in the brain are connected to all the other neurons so second thing is uh neurons uh use event based communication uh which means that communication really happens only through discrete events that send messages between units and so it's not

the case that each neuron is talking with the next neuron all the time right which is uh on the other hand most of our artificial neural networks really do work like that so there's like an output that goes to the next unit or the connected units pretty much all the time uh the third thing and again this is because of just uh very practical constraints in biology is that learning is local in time and space so you cannot uh have uh altw communication between like neurons in the brain to do learning because it's just impractical and uh so what this means is that each synapse has to use information that's available mostly locally to the synapse and then with like very very low dimensional Global signals incorporated into this so uh and and and so this this is par in uh is it's local in time as well and the fourth aspect is uh temp oral coding so because time is available in some sense for free quote unquote uh neurons can use time to compute they can use time to actually send messages uh and it can be used in various different ways uh so you can neurons are able to trade off computation for time in that sense there are lots of other interesting properties asynchrony for example uh and then also some sort of like distributed learning that happens in various parts of the brain so I won't be talk going into detail into any of those at the moment okay so let's come back to this plot which showed the scaling of various things and so now I've added back the labels from the original plot and so uh this Gray Line over here is the dense 2D silicon scaling that we have and so this middle line over here is what would happen if you actually took the silicon and put it in three dimensions but without sparity and so you you obviously do get like better scaling but there are some practical challenges that need to be overcome to actually have three-dimensional silicon uh mostly related to heating issues and uh the biological brains are all in this regime of sparse three-dimensional um substrates right so this sparity is really what makes the difference uh I mean given you have a three dimensional substrate sparity is really what makes a difference in uh allowing this much better scaling property of these U computational devices as it were so this really means that we need uh apart from uh the right algorithms we also need the right hardware and uh there has been quite a lot of work on biologically inspired Hardware that can make use of the SP asynchronous communicating neural network units uh so uh and this is kind of like this illustration uh that Intel has so they they've been like also working on one of the uh neuromorphic Pro processors called lii and uh this is like a really great illustration of like the differences between conventional Computing and neuromorphic Computing but essentially in conventional Computing you have synchronous clocking you have sequential threads of control whereas in neuromorphic Computing you have completely asynchronous even based spikes and parallel and sparse compute and uh it's also I mean so

gpus are kind of in between so which allows parallel Computing but it's still synchronous clocking and it's kind of more optimized for dense computation and uh so these neuromorphic chips are essentially chips that use these biological um ideas and are able to work with sparsity in both time and space and also able to work with asynchronous computation and so the state of the neuromorphic chips right now just to kind of uh summarize uh what's what's been happening so I mean I I personally don't work on the hardware aspect of it but mostly in uh the algorithms but there has been recable amount of progress with neuromorphic chips so for example um in 2021 at least uh so this is like a plot which shows U the uh relative performance of various uh devices in comparison with this Intel Loi chip which is a neuromorphic device and uh so if a device is if a point is above this dotted line it means that um Loi the neuromorphic device does better and if it's below it means the conventional device does better and all of the points here are conventional devices so you have a CP you have CPUs gpus um some kind of like accelerator deep learning accelerator and IBM True North which is also a neuromorphic device but a little bit older and uh the bat size also makes a huge difference if you have bat size of one the neuromorphic devices tend to do better in general and larger bat sizes uh gpus T to do better and so uh so so what you see over here is uh well first of all uh there are certain like biologically inspired models which are these blue ones over here that do much better on a neuromorphic device compared to uh conventional Hardware on the other hand uh what you see is that even for some of these biologically inspired uh models if you start using large patch sizes so if you look at this purple Square over here uh it uses something called an LSN which was one of the uh papers that I was involved in uh a few years ago and if you combine that combine that with deep neural networks for a bat size of one uh the neuromorphic device does better but then if you improve increase the batch sizes then uh gpus suddenly start doing better so uh so neuromorphic devices are still not there yet it's still work in progress and uh Loi for example is a digital neuromorphic device and there's lot of uh variety of possible neuromorphic devices out there a lot of people working on very cool interesting ideas uh but there is this promise of uh uh the neuromorphic devices having really intrinsically better scaling curves if the engineering is done right okay so now uh I'm going to talk about algorithms a little bit so let's just go back to this slide uh where we looked at what biology does uh that uh I think is important for scaling up these models and uh one of the initial kind of attempts at using some of these principles was what are called spiking neural networks and uh spiking neural networks are essentially a direct biological uh I mean it's it's a model of a neuron that is really directly uh derived from biology so as I said in

biology you have neurons and then if you kind of stick electrode in the neuron and measure um uh measure what it's doing uh what you see is that it has this action potential right so it's suddenly there is this surge uh this the sudden change in like the potential and I'll tell you what this potential is in a minute and uh but this uh action potential as it's called is a spike and uh that's what is used for communicating and most of the rest of the time the neuron doesn't do much and so this action potential is essentially uh because the neuron is constructed in such a way that there is a voltage potential difference between the inside and outside of the neuron and most of the U and this potential is kind of uh constructed through uh the movement of ions the neurons control the movement of ions but across the membrane across the cell membrane and which kind of creates a membrane potential and this is used to kind of trigger the action potential of a neuron which leads to ionic currents inside the neuron which triggers the synapse and then there's a chemical messaging across the synapse which goes to the next neuron and so on so that's kind of like the biological process and this can be simplified into like like a circuit uh if you will an RC circuit and then it can further be simplified into a computational model where uh there are inputs coming in as spikes which are just zero or one values and the neuron does a weighted sum of all the inputs coming in so this is kind of very similar to what artificial neural networks do except that in the case of spiking neural networks the inputs are binary and very very sparse in and uh so the neuron kind of integrates this and then if the internal state of the neuron goes above certain threshold then it also emits a sparse discrete Spike and uh so if you uh kind of uh look at the activity of the neuron this is what it looks like so the membrane potential if if there is a input that looks like this so some sort of uh constant input then the membrane potential goes up every time a Spike arrives at the cell body so in this case just the neuron the computational neuron and whenever this membrane potential goes above a certain threshold then the neuron emits a spike right okay so this is a very very simplified model and this model is called The Leaky integrated and fire model U so you can kind of write this down in terms of equations and essentially uh uh okay I mean so I I don't think I will go through this but um maybe the important thing to kind of note is that uh there is some sort of uh the output of the neuron which is Z over here is one if the internal membrane potential is above threshold and zero if it is not and this Z is the only thing that other neurons see so the membrane potential is completely internal to this neuron and no no other part of the model has access to it uh but this one zero value this binary value the spike that's emitted is what uh the other neurons see and this this kind of depends on the threshold okay so now uh spiking neurons are great because you have this temporal sparsity

now right so you're not like sending messages all the time but you're only sending messages every now and then but uh you need to train these models and in a standard RNN you can just use back propagation Through Time by defining a loss function and then calculating the gradients of the loss function back propagate It Through Time into the model and then update the the weights based on this gradient but the problem with spiking neural networks is that the spike emission function is non- differentiable which means that it's not so straightforward to just apply back propagation Through Time directly because that depends on all the parts of the model being differentiable but there is kind of like this uh very crude trick as it were that works surprisingly well uh that we discovered uh back back in 2018 along with colleagues which is that you can I mean so this uh blue curve over here is the spike emission function which is this kind of thres uh step function which where the derivative is not defined at the threshold and uh we kind of realized and I mean a lot of people realized at around the same time that you could just Define a pseudo derivative so you could Define a derivative at that point quite AR Lily and uh within within some constraints if you uh so for example we uh defined it in this uh P linear function and uh it tends to work really really well so you just replace the derivatives of this uh non differentiable function at that particular point with the value that's derived from the pseudo derivative and then you can just train all these models with back propagation through time and they work work really great but turns out that that's not enough I mean so if you have spiking neuron and use back propagation through time uh that's really not enough to get the same level of performance as artificial neur networks because the second aspect of recurrent networks that we needed was long short-term memory and recurrent networks uh need to have some sort of memory by default U so just standard artificial uh recurrent networks and leaky integrated fire neurons neither of them have this long shortterm memory so in the uh deep Learning Community people tended to use this model called LSM uh which was really really powerful because it had this specific mechanism for storing and updating a me something called a memory cell so it could U gate the input coming into the memory cell it could uh decide to forget things it could also gate the things going out of the memory cell and this really helped uh this model to work really work have have long memory essentially long short-term memory and so we wanted to do something very similar for spiking neural networks because leaky integrated fire neurons by default do not have uh this memory and so one of the interesting observations uh that we had was that in Neuroscience the activity of the neurons don't really correspond to like a very typical leaky integrated fire Neuron model pi because there seems to be some sort of adaptation happening

so if the neuron has constant input so I mean this is this is kind of an example where if if you give the neuron constant input for leaky integrated fire neuron you mostly expect the spikes to be evenly spaced U because every time it hits the threshold uh the neuron spikes and then it's reset to a lower value and then the input kind of drives the mem brain potential up again it spikes again and so on but what you actually see in biology is some sort of adaptation so if a neuron spikes then after every Spike the next Spike uh is pushed back it it spikes less for a while and then after that uh it spikes again and then again it spikes less for a while and SW on there's kind of like adaptation of the Dynamics uh and you see this everywhere so in the pyramidal cells in the mouse and in the human neocortex so this is data from the Allen Institute where they kind of characterize cells from various different animals and um so this kind of measure of this adaptation is called an adaptation index and essentially what you see is there's a big chunk of neurons in uh humans and mice which are which have an adaptation index that is large which means that there is this adaptation happening and so what we did uh was to try to model this uh and include this F and leaky integrated neurons and we came up with a model called uh an Adaptive leaky integrated INF fire neuron where the idea was quite simple it was uh to adapt the threshold every time a spike happened so every time a spike happened the threshold would go up a little bit and then DEC back to the base value and so as it turned out this simple mechanism so this this kind of threshold adaptation was enough to uh put the performance of spiking neurons on like very very close to lstms when trained with back propagation through time so this is a plot where you see this uh so this was uh tested on mnist a sequential Mist where you give the uh input one pixel at a time to the recurrent Network and then it actually to classify what the digit was at the end and uh so just standard leaky integrated fire neurons does really really poorly so it has like 50% test accuracy whereas just adding this adaptive threshold which is the model that we called LSN gets the performance up to 93.7 right and lstms are up here at 98% and um so with with some other additions uh and using parameter sparsity uh it goes up to 96.4 right and this is with spiking neurons so it turns out that uh so this this was like one one example of like using like biological data and basing the model on that and seeing a huge Improvement in what the models could do okay uh so how am I doing on time uh uh so we're about halfway through the session now so how long do you much more do you think you'd like um probably about 15 more minutes is that okay we're halfway through the hour H you can take an extra 15 more minutes it means we'll have sort of 15 minutes for chat at the end so it's up to you kind of how you want to uh divide it I would like to remind everybody if questions they can kind of jump

in as we go as well there's it's it's really dense this one I'm loving it but like I I guess yeah the timing we might have a little bit less time for chat at the end but if people want to stay on um the call like I can stay on as well for a bit longer if you can as well and and yeah absolutely sure I mean yeah so I will I will try to like uh do speed it up a little bit at least like skip some parts and then you can go and look up the paper right okay don't go all the details okay but uh so uh so we kind of also had um um models that were able to take advantage of this temporal coding that I mentioned actually using time to do computation and uh so to kind of quickly uh summarized this the idea was that so in the brain uh you kind of have a specific mechanism that controls the speed of transmission of signals uh so this is called um um so there's this thing called a myelin sheath and there's there's been like a lot of recent evidence that uh so the myelin sheath is has been known to control the speed of the signal transmission there's been a lot of recent evidence that the myelin sheath actually adapts in adults which means that that seems to be like an additional dimension of learning that happens in the biological brain in adult brains and um so we wanted to kind of see what can be done with just controlling the speed of transmission so which in a computational model translates to a delay so a delay in communication between one neuron and another neuron and so what we tried to do was to test out uh if we could train a model with only synaptic delays uh without training the weights at all and see how far we could go on deep learning benchmarks right and so we did this and uh so we had like a temporal encoding and we constrained the weights to only three values so some so it was something like 10^{-1} and 10^1 uh and the weights were randomly initialized and then fixed and then we trained only the delays of the uh Network synaptic delays and it was quite interesting to see that just by training delays you can get performance that's comparable to like your standard uh both both uh standard artificial neural networks uh without convolutions and uh standard spiking neural networks uh where the weights are trained can I just ask is that with the Adaptive neurons as well that you were just talking about or is this just with the delay uh so this does have some form of adaptation in the neuron uh so it's actually a slightly different form but in this particular case uh we were working with feed forward networks and not recurrent networks okay so uh the memory didn't matter so much okay thank MH so this is an example of like using uh time uh for computation and we there's ongoing work to kind of extend this even further and to see really how far we can uh push using time as to compute because so what is interesting is uh when when you use delays for example uh in a computational model if it's implemented properly in appropriate Hardware a delay is just uh a neuron not doing anything until a certain point right which means that when

you're when it's not really doing anything when it's not sending um message to the following neurons you're not consuming any energy and so this is a very uh nice way to make the models more efficient by tra by having delays and uh by using by kind of like reducing the activity of the overall Network okay so uh so so far we've been like looking at at spiking neural network so which were these models that were based kind of like very very closely based on biological Dynamics and uh still using all of these principles of sparsity and event based communication and temporal coding but spiking neural networks of this particular form still don't work um as well as the best deep uh Network models so essentially in 2018 lstms were when when we were doing this adap neurons lsms were the dominant model but then as we as I mentioned in the beginning of the talk uh there have been like newer models that have been performing even even better and so now we really have these Transformers and related models that do really really well and scale really really well and spiking neural networks pretty soon kind of hit a uh ceiling in terms of how well they could do and we had to think about uh ways to push the effectiveness of these models beyond that while keeping all the advantages in terms of sparsity and efficiency so we started thinking about uh really using these principles biological principles and designing models uh from first principles uh to take advantage of sparsity and event driven Computing and so on but then without really modeling the neuron directly based on biology uh and the I mean in for the sparsity in time the basic idea really was that you had like this threshold gating function uh that would decide when communication went uh to other neurons and when it did not and we realized that we could just put this use this for other uh architectures that tended to work really really well for things like language modeling so uh and and with both uh these models and with spiking Dr networks what happens is that uh because you have this uh activity sparsity which is essentially that not all the units are active all the time when you train these models the backward pass is also sparse because you only really need to propagate the gradients through the uh units that were active and a few other units because of the particular shape of the pseudo derivative that we chose but you do not have to propagate the gradients through all the units and if you can actually take have some Hardware that can take advantage of this you end up getting like way more efficient training as well so not just the forward bus so this is kind of uh uh an illustration of this so the forward pass over here with these blue arrows is uh only goes through units that are active and the backward pass uh when you're calculating the gradients again only goes to the units that are active and a couple of other units uh that were close to being active but most of the other units are skipped entirely in this computation and we decided to to take this model called a gated recurrent unit

which is another recurrent Network model that works uh quite well I mean it's still lot on par with Transformers but uh it works uh so mean it was it was one of these models that had like a really quantitative advantage over leaky integrated Fire based spiking networks and we wanted to make it activity sparse as well and uh so we took the Gru and added a even generating mechanism to it using a gate and uh few a few other features like having a reset and having um refractory period and so on and with this model we were actually quite easily able to match the performance of lstms or or like the best language models recurrent language models u in terms of perplexity so that's that's kind of like what you see over here so there are like two different data sets one is the penry penry bank and one is the wikitex 2 and the uh bold values are like the best recurrent Network model performance on these two data sets and with our uh event based Gru model as we called it we were able to get um perplexities that were very very close to these um recurrent models but we had extremely high activity sparsity right so if you look at just maybe this one example uh with the egpu on the penry bank we had about 80 85% activity sparsity and so if you had had Hardware that could actually take advantage of it it would uh it would use like 85% less computation if the hardware could actually take full advantage of it and the backwards parity was close to 50% so you really end up doing like 50% less computation for training the same model right and we were also able to use like way fewer parameters uh to do this so overall uh these models I mean so the thing is this these resarch that we showed uh all the training was done on gpus because we actually don't have the hardware that can take advantage of all of this sparsity uh but this is uh in some sense like calculation of exactly how much energy we would be able to save if we had this hardware and U so we we were we were also kind of able to tune uh kind of tradeoff uh sparity with performance within certain limit so you see here that we don't even though we are very close to the recurrent model uh performance we don't actually reach it and then there is a way to kind of make it less spse and reach this uh close and you still kind of end up with some Energy savings uh so for example uh so we actually implemented this on a neuromorphic chip uh recently uh is a neuromorphic chip called Spiner 2 which is developed by uh some folks at T Reston in Germany and what was kind of interesting is uh in if you if the energy consumption for single batch uh for Spiner 2 which is this green bar over here as you can see is significantly lower than running it on a GPU so uh this blue bar is the a100 GPU which is like the one of the more recent ones and the orange one is like a slightly older GPU so it's like a little bit slower and uh Spiner 2 and this was like actually a very uh Proto uh the implementation was essentially a prototype so we were not actually making full use of the chip but uh it was able

to make use of some of the sparsity um on the device and you kind of get like a huge energy Advantage even with that although for larger batches you don't right so this is kind of still one of the uh difficult areas of neuromorphic computing where it's not able to take advantage of larger batches so um yeah so this is kind of uh on archive you can go look look it up uh but uh so we kind of like getting there slowly um getting getting all of the sparsities on Hardware okay so we can also use sparsity to make training very efficient I won't go into this in detail but essentially you can use algorithms that so far were considered completely uh computationally prohibitive uh in terms of the computational expense you can actually that so these algorithms uh become more practical if there is activity sparsity and parameter sparsity okay so so and we we have explored like a really really really tiny part of the properties that can make model scalable which is just activity sparsity and parameter sparsity and we have mostly explored it only in terms of algorithms and so there's obviously like a long way to go before we start reaching this scaling uh of biological uh substrates and biological algorithms but we're kind of like trying to move towards this direction so that's kind of like the conclusion which is that um sparsity and locality are essential for scaling up models due to energy constraints and there might there are also other clever ways to use physical substrates uh to use to improve Energy Efficiency so using time to compute communication and others and the next step is to really like try to scale it up and start to catch up with uh Transformer based models on tasks that are a lot more closer to what people use in the real world so a lot of this work was done with lots of other people uh so this is the group that uh group and collaborators that we have we have like a large group in ruer University bohome so you might you some of you might have actually seen a talk by David Capel who gave a talk at mream a while ago so uh and we have a few people from T Tron and uh there's is the group that actually develops the Spinner to hardware and makes all of this possible so thank you very much um yeah hopefully we have some time for discussions thank you so much that was absolutely fascinating there's a lot to unpack there so Harris you've got your hand up if you want to ask a question yeah ER I just wanted to uh say Kudos it was very interesting and I have a couple of uh questions related to this presentation uh first of all H you you poed one um very interesting question that has to do uh with the following um can we scale even better let's say moving effectively moving beyond Transformers and as far as I understand it uh because uh these large language models the llms are very are all the hype these days so as far as I understand it not all llms are are Transformers uh but the other way around is true and the reason for for that H is because uh there are some um llms that are not based in in Transformers as you mentioned uh you

mention the recurrent neural networks of course and that's one one type let's say and uh another uh uh subtype that is based on rnet is the long shortterm memory networks that you are group is actually investigating and uh what I wanted to say is that trying to uh put everything together is there a possibility for moving beyond uh the Transformers and scaling even better by opting for what is called poly Computing and and I mean poly Computing that is capable to uh to combine uh different types of computing in a multi Computing Paradigm like classical Computing Quantum comp Computing neuromorphic and even Optical so what do you think about that yeah that's a very interesting question I mean so there's definitely like a lot of potential in these unconventional Computing devices that various people are developing right and each of them have different strengths and different weaknesses so it would make sense to combine them together in a way that uh we use the strengths of like some of them without uh being affected by their weakness so much but uh well I mean at least in terms of efficiency itself if using like multiple devices or multiple different uh Computing nodes always kind of leads to this problem where you do a lot of communication and communication is expensive right so even on a CPU or a GPU or like most Computing devices that we have today uh the communication is really the bottleneck so that's probably uh I mean that's actually like one of the main reasons why we are looking at activity sparsity because that's that has to do with communication so communication tends to take a lot of energy it consumes a lot of time and so uh when with like multiple different types of commun computing the main problem that would have that would need to be solved is how do you make communication efficient both like on a physical substrate you know like both both in terms of like the device itself and also in terms of algorithms okay thank you thank you very much um yeah I had a couple of questions as well I think it's it's fascinating um that you're saying essentially that you can get get the same performance of the lstms but like with about 80% less power like I think that's really amazing do you think that this these types of architectures will be used sort of in the near future to make say gpus more efficient or or implemented in some specific use cases maybe where sort of the sort of temporal sparsity is particularly important the temporal coding is particularly important so I'm quite interested to hear your take console what the applications where you see this being used in in the near future MH so the temporal coding I mean I've not really seen uh much attempts to make use of it I think it's it's probably like one of those things which is like the hardest to make use of both both from algorithms and Hardware point of view but people are trying to make use of uh this activity sparsity so Nvidia for example recently uh added support for a certain type of sparsity

like it's still like very very limited but uh they started adding support for a certain type of sparsity which in principle could be used for activities parity quite well and there have also been like a couple of deep learning accelerator companies that uh do activities or I mean so okay I mean so actually like the I mean there's probably like a bit of context that's required so the reason that activity sparsity is really hard to make use of in Hardware is because it's Dynamic right so if you have parameter sparsity and if the parameter sparsity is fixed you kind of know which uh particular weights you need to load at what times and which ones you don't but if this is changing all the time then it's really really hard to optimize uh its use and so this Dynamic sparsity is what is like extremely challenging for gpus and like most conventional devices so uh nvidia's attempt at adding this support for sparsity is to to some extent address uh being able to use Dynamic sparsity although in's case you cannot use uh I mean so basically using even static sparsity is very very hard unless it's designed in like a very specific way uh so and and so this deep learning accelerator company that I was mentioning so they really have done they've designed a chip completely from scratch so it's a company called cerebras which can make use of Dynamics sparsity and uh and it's also like super scalable because essentially what they've done is created like this huge chip on a single wafer which means that the communication overhead is much much lower and uh it works like really really great for training like extremely large models if if you have the right kind of sparsity on them right so it's still not like completely general purpose it's still like very closely tied to Transformer architectures but it can use activity sparsity really well uh if implemented correctly but uh they don't really go beyond that so they don't do like time based Computing or make use of like any of the other aspects especially asynchronous Compu Computing for example is as far as I know only neuromorphic people working on neuromorphic devices seem to even use it although well I mean I guess like somewhat ironically we haven't fully figured out how to use it effectively interest I was actually just going to ask you a bit about Dynamics vast because I'm in interested in um sort of neuro neur taking inspiration from neurod development and I I think it's really interesting that in the developing brain like say from about Age 2 to 10 you have huge amounts of synaptic pruning like you actually lose about 50% of your synapses and that's essentially a key component of how we learn um so before I saw this talk I was thinking about how if anyone's actually taking inspiration from neurod development for example thinking about sparsity and how the brain sort of develops by pruning away all the sort of unwanted connections is is that which is people are thinking about does that fit into what you were just saying a little bit about Dynamics B here is that slightly

different uh no I mean it's actually very very closely related and uh I mean we've been like thinking about it a little bit also but so essentially uh what we have come to realize is that so there are two different aspects to having a model that performs well so one is just the inference so if you have sparsity uh you can make predictions with this model or like kind of run the model forward uh and uh any sort of sparsity helps it makes it very very efficient but this does not necessarily make the model easy to train right so you need to have there is this like really orthogonal uh concern where you need to be able to train the model well and it's little bit in conflict with having sparsity so like one of the well at least like from a very very non-expert perspective what it seems like is this neuro development uh this kind of like pruning that happens it seems like there might be like a learning phase where a certain type of learning happens that is a lot harder to do if the overall network was really sparse and then you have this like huge pruning space face uh and then you have like a different network that learns that can still learn but like a lot more slowly perhaps or but but it's like a lot more energy efficient it interesting because yeah like the brain although it's pruning it's also growing and and the part of the reason why it's growing so much is because the myelination which is interesting because you talk about that in terms of the synaptic delays so maybe sort of there's some balance between the sparsity pruning and then sort of the delay marination that's introduced somewhere in a dynamic way um that's important to do I don't know yeah I mean I think uh I mean yeah so I mean obviously like in biology uh you know like specific things don't just do one thing right they do like a number of different things so that's that's the case with Malin as well for sure uh but uh so kind of what what is interesting is that with delays I mean if you can control the delays and if you can work with longer delays then you actually need less activity in the network because uh I mean so if you take like a network with like absolutely no long-term memory so like Ley integrated F neuron or a v Network and you kind of try to train this network to store something for a longer time one of the ways that it can discover to store things is literally uh persistent activity so you have like one neuron that keeps firing all the time because it's storing one particular bit of information and then at the end if this neuron is firing you know that bit is stored and if not it's not right so it's kind of I mean it doesn't always emerge and it's pretty hard there's a reason why these R networks don't work well for like long-term memory but uh if you if you make the problem simple enough you do see this right and in when you train like a spiking Network or an lstm you do see like persistent activity on and off for various things and if you had delays you could just completely avoid this right you could just like uh say this neuron will will send the message when it's ready after like a few

time steps or so and only then the other neuron needs to process this information okay interesting that might touch on my which I'll make my final question because we have to drop and everyone's dropping off absolutely unshare your screen sorry I should have probably asked to do that earlier um I'm I'm quite interested in this idea of so artificial neural networks they have the sort of this like always on an activation function whereas people contrast that with spiking neuron networks which are like neurons and like they're on or they're off and what's interesting I guess from a neuroscience perspective is because a lot of the brain is actually gl cells which actually do have more this continuous Activation so the brain is actually quite a hybrid between more like ANNs and SNNs essentially so I'm super interested in sort of if if you think um like maybe the future will be um hybrid in some way um oh Carl Frist has entered the way waiting room I think I'll admit him I don't know if he's in the right meeting we're meant to have an SA meeting okay maybe yeah so that was my last and final question sort of do you think the hybrid system um would uh be something which people are interested in or looking at in the future or is that not something that I see uh that's an interesting question so well I mean the qu I mean so I think the related question that needs to be answered is do hybrid systems really provide any sort of advantage in terms of uh computational Power or any any other thing and at least like as far as I know there is no reason I mean so if you can get like this uh event based system to work and at least like theoretically it's they're like equivalent they're supposed to be equivalent to artificial like you know non-event based systems then there wouldn't be any reason to use a hybrid system but it could very well be the case that there are certain constraints where just an event based system would be a way to slow or you know so that that might be like a concern so it just like takes too long to reach decisions or something like that and then and then you might need like hybrid systems as well yeah I mean so I'm happy to discuss I mean if you're interested in this area I'm happy to discuss with you offline as well I'm super interesting because as well I like the sort of you get with the gl cells there actually some electrical communication they have Gap Junctions and they have these sort of slow oscillations which almost like event based oscillations almost take advantage of or ride so there's sort of an embedding nesting there of the the uh spiking and non-spiking so um yeah it's something I'm super interesting I don't I just yeah I don't know if the Silicon world is as interested in it as me though as a neuroscientist yeah but I'm happy to chit chat to it offl okay well thank you so much that was absolutely fascinating um I'm sure people might want to reach out to you with some questions so of offline because this will be put on YouTube so if they do what would be the best way for them to contact you if that's okay

yeah absolutely it's okay I mean I have my contact information on my website okay great so email is probably like the best way to contact me cool okay right thank you so much and that's everything that I have to cover um yeah thank you thank you very much for inviting me to the talk it was it was very interesting yeah amazing yeah I'd love to stay in touch and chat some more about this as well so thank you so much absolutely all right see you byebye